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BTech Cse, 3rd year

sec:"B"

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ROll no. = 11

Experiment No. 2

Experiment Name

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Aim

To perform Exploratory Data Analysis (EDA) on a real-world dataset using Python in order to understand the dataset's structure, identify trends, detect anomalies, analyze feature relationships, and generate meaningful business insights through descriptive statistics and visualizations.

GITHUB LINK: https://github.com/Arpitcu24250181/Mastery-in-Data-Analytics-and-Visualizations-and-Career- Advancement/blob/main/Experiment%202.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
df = pd.read_csv(next(iter(uploaded)))

print("Dataset loaded successfully!")
```

```
Dataset loaded successfully!
```

```
df.head()
```

	Order Date	Region	Category	Sub-Category	Product Name	Sales	Profit	Discount
	2024-01-05		Technology	Phones	Smartphone A	850	170	0.10
0	2024-01-12	West	Furniture	Chairs	Office Chair	420	55	0.20
1	2024-01-18	East	Office Supplies	Binders	Binder Set	75	18	0.05
2	2024-02-03	Central	Technology	Accessories	Wireless Mouse	180	42	0.10
3	2024-02-14	South	Furniture	Tables	Conference Table	950	95	0.30
4		West						

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 30
Number of columns: 8
```

```
print("Column names:")
print(df.columns.tolist())
```

```
Column names:
['Order Date', 'Region', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Profit', 'Discount']
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
```

```
Data columns (total 8 columns):
 #   Column        Non-Null Count Dtype  
---  ------        -------------- -----  
 0   Order Date     30 non-null    object 
 1   Region         30 non-null    object 
 2   Category       30 non-null    object 
 3   Sub-Category 30 non-null      object 
 4   Product Name 30 non-null      object 
 5   Sales          30  non-null   int64  
 6   Profit         30  non-null   int64  
 7   Discount       30 non-null    float64
dtypes: float64(1), int64(2), object(5)
memory usage: 2.0+ KB
```

```
df.describe()
```

	Sales	Profit	Discount
	30.000000	30.000000	30.000000
count	577.500000	104.066667	0.133333
mean	453.145615	83.364978	0.079148
std	75.000000	18.000000	0.050000
min	165.000000	42.750000	0.050000
25%	535.000000	77.500000	0.100000
50%	832.500000	135.000000	0.200000
75%	1850.000000	330.000000	0.300000
max			

```
df.describe(include='object')
```

	Order Date	Region	Category	Sub-Category	Product Name
	30	30	30	30	30
count	30	4	3	10	30
unique	2024-01-05	West	Technology	Phones	Smartphone A
top	1	8	10	4	1
freq					

```
for column in df.select_dtypes(include='object').columns:
    print("\n", column)
    print(df[column].value_counts().head())
```

```
 Order Date
Order  Date
2024-01-05    1
2024-01-12    1
2024-01-18    1
2024-02-03    1
2024-02-14    1
Name: count, dtype: int64

  Region
Region
West       8
East       8
Central    7
South      7
Name: count, dtype: int64

 Category
Category
Technology         10
Furniture          10
Office Supplies    10
Name: count, dtype: int64

 Sub-Category
Sub-Category
Phones         4
Chairs         4
Binders        3
Accessories    3
Tables         3
Name: count, dtype: int64

 Product Name
Product  Name
Smartphone A        1
Office Chair        1
```

```
Binder  Set  1  Wireless
Mouse  1  Conference  Table
1
Name: count, dtype: int64
```

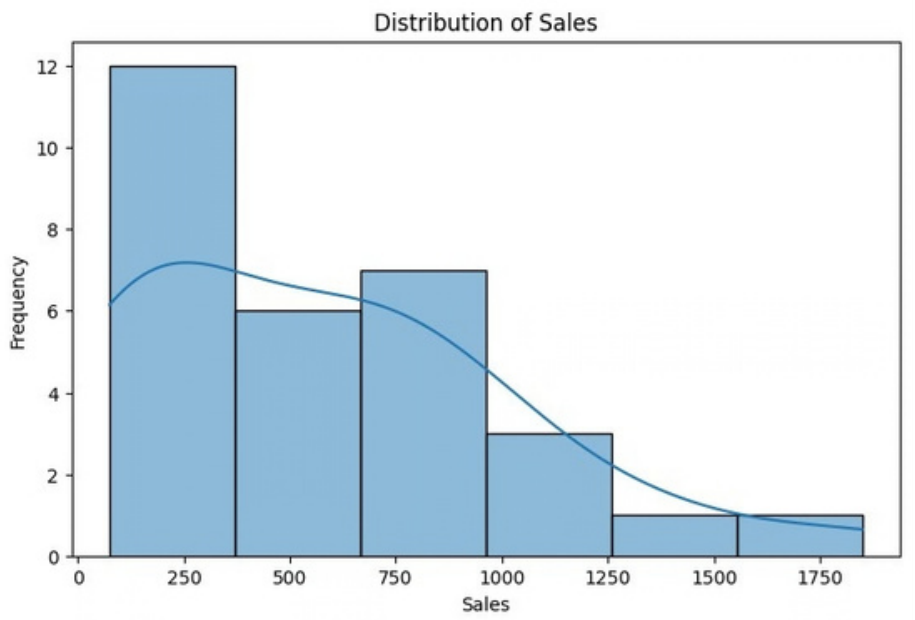
```
print("Missing values:")
print(df.isnull().sum())
```

```
Missing values:
Order      Date 0
Region          0
Category        0
Sub-Category    0
Product Name    0
Sales           0
Profit Discount 0
dtype: int64    0
```

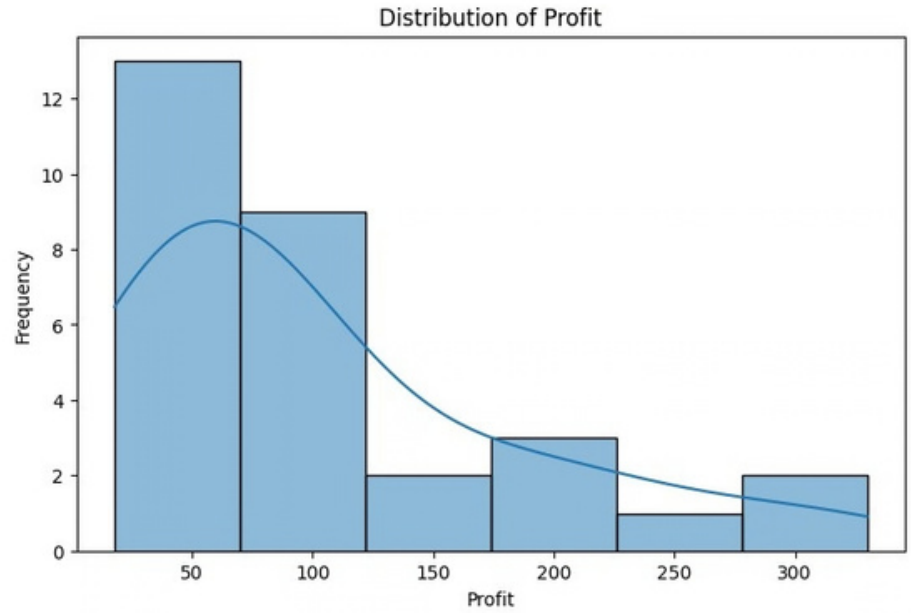
```
plt.figure(figsize=(10, 5))
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values Heatmap")
plt.show()
```



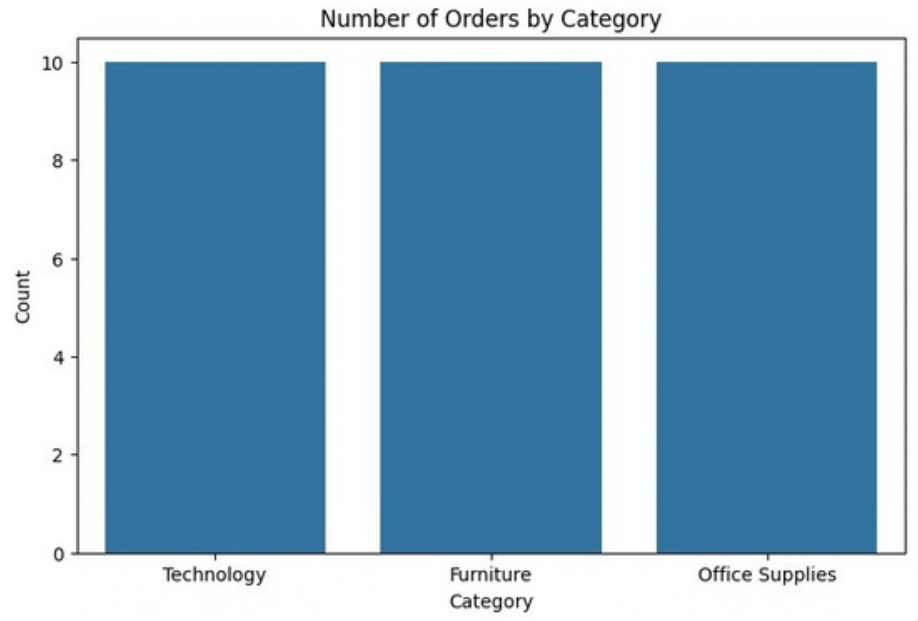

```
plt.figure(figsize=(8, 5))
sns.histplot(df['Sales'], kde=True)
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```



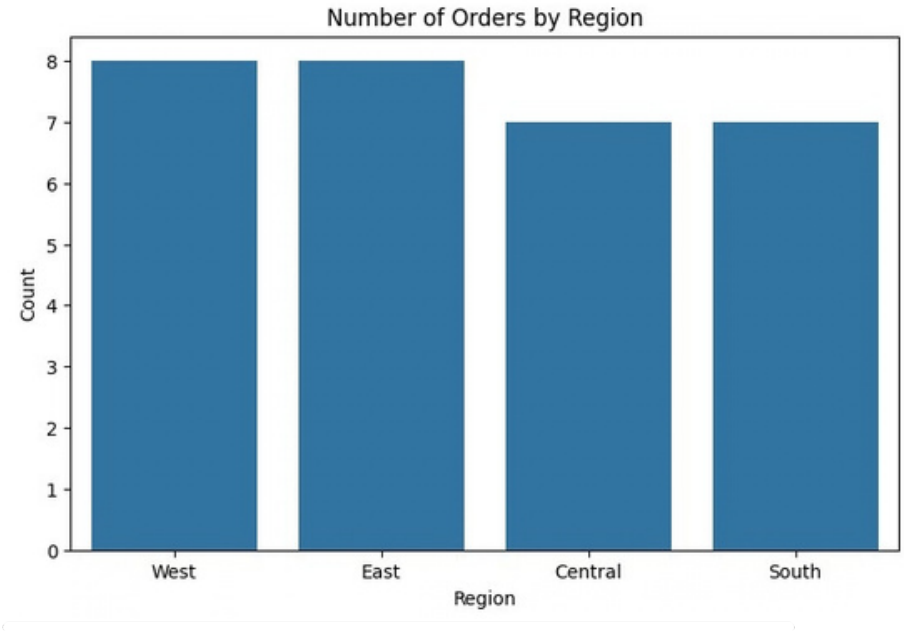
```
plt.figure(figsize=(8, 5))
sns.histplot(df['Profit'], kde=True)
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```



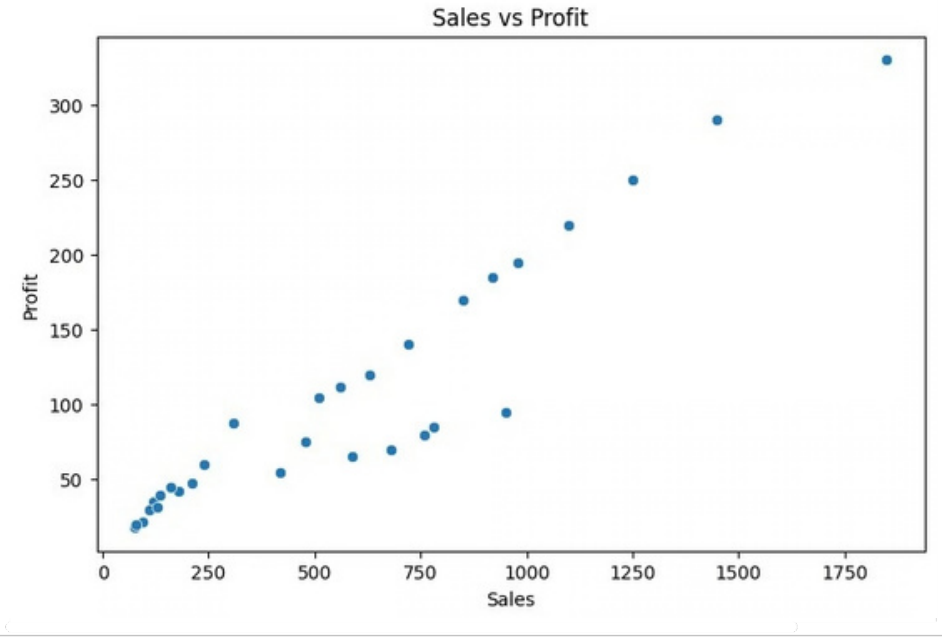

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Category')
plt.title("Number of Orders by Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```




```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Region')
plt.title("Number of Orders by Region")
plt.xlabel("Region")
plt.ylabel("Count")
plt.show()
```




```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='Sales', y='Profit')
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```



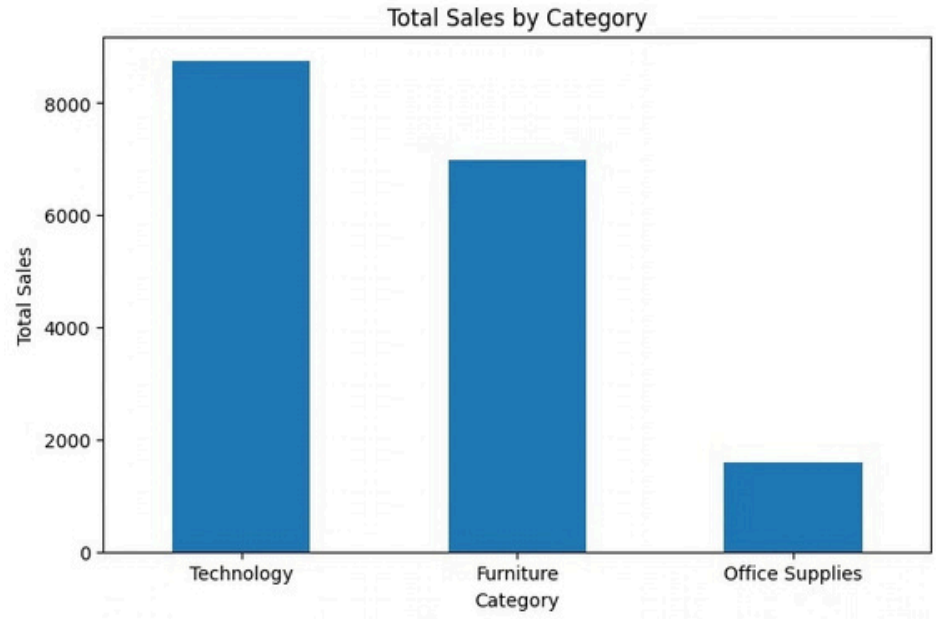

```
category_sales = df.groupby('Category')['Sales'].sum().sort_values(ascending=False)

print(category_sales)
```

```
Category
Technology         8750
Furniture          6980
Office Supplies    1595
Name: Sales, dtype: int64
```

```
category_sales.plot(kind='bar', figsize=(8, 5))

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()
```



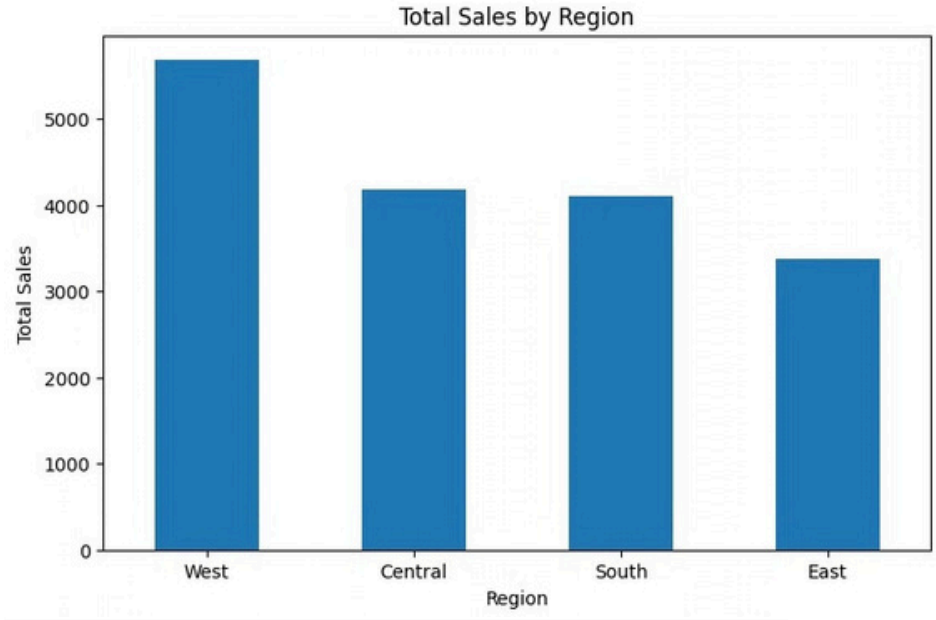

```
region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)

print(region_sales)
```

```
Region
West       5680
Central    4175
South      4095
East       3375
Name: Sales, dtype: int64
```

```
region_sales.plot(kind='bar', figsize=(8, 5))

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()
```



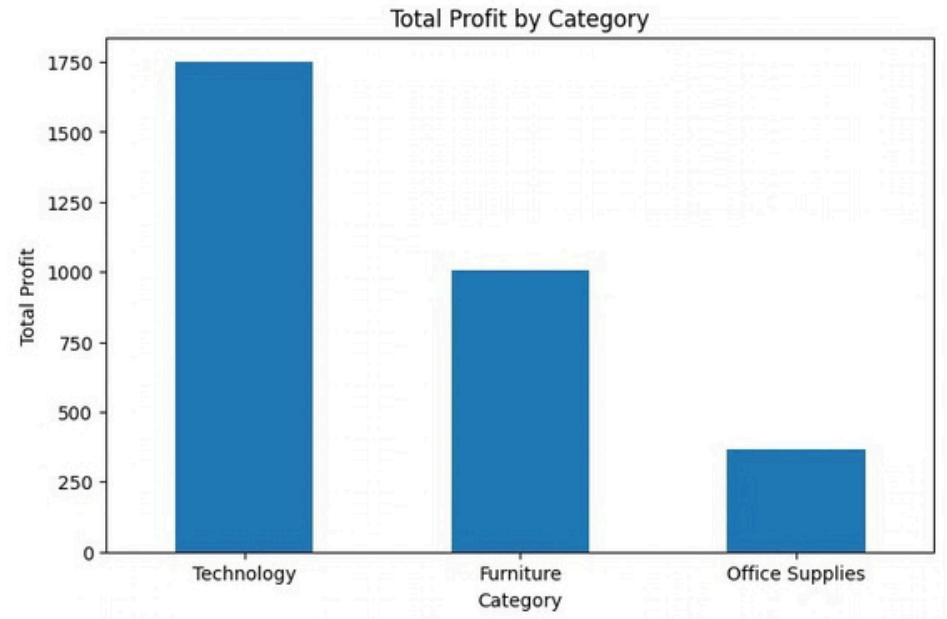

```
category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascending=False)

print(category_profit)
```

```
Category
Technology         1750
Furniture          1007
Office Supplies     365
Name: Profit, dtype: int64
```

```
category_profit.plot(kind='bar', figsize=(8, 5))

plt.title("Total Profit by Category")
plt.xlabel("Category")
plt.ylabel("Total Profit")
plt.xticks(rotation=0)
plt.show()
```




```
numeric_df = df.select_dtypes(include=np.number)

correlation = numeric_df.corr()

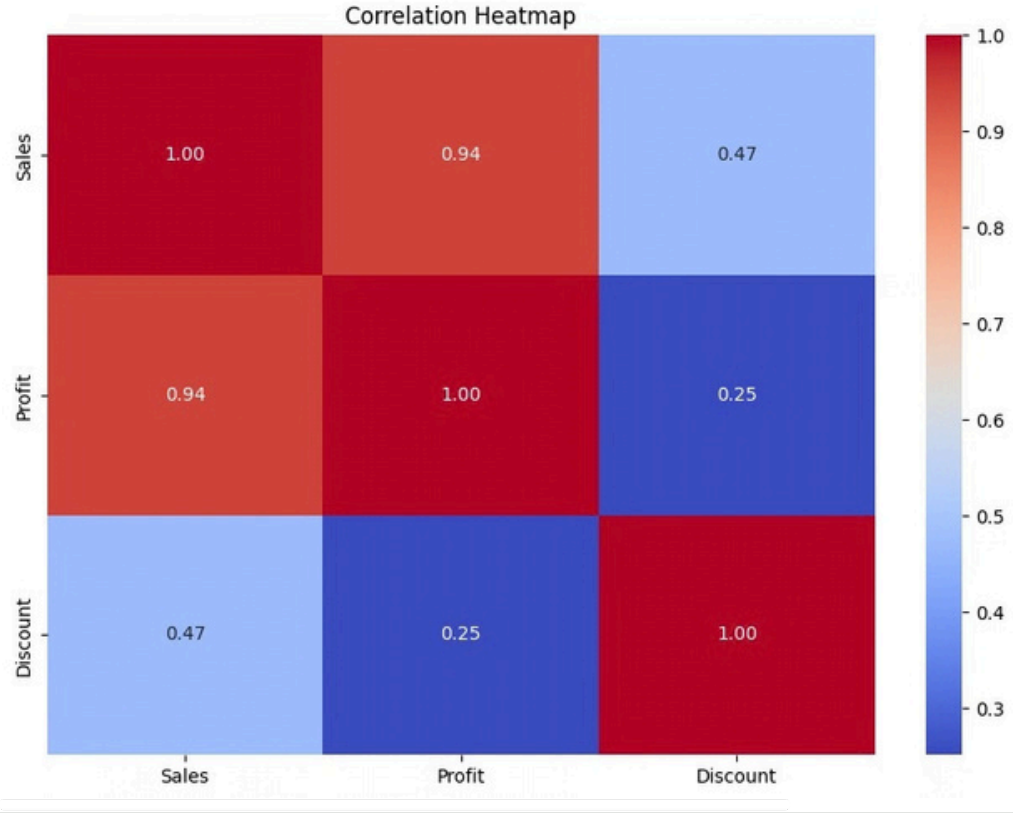
print(correlation)
```

```
             Sales      Profit Discount
Sales        1.000000 0.942052 0.471830
Profit       0.942052 1.000000 0.251028
    Discount 0.471830 0.251028 1.000000
```

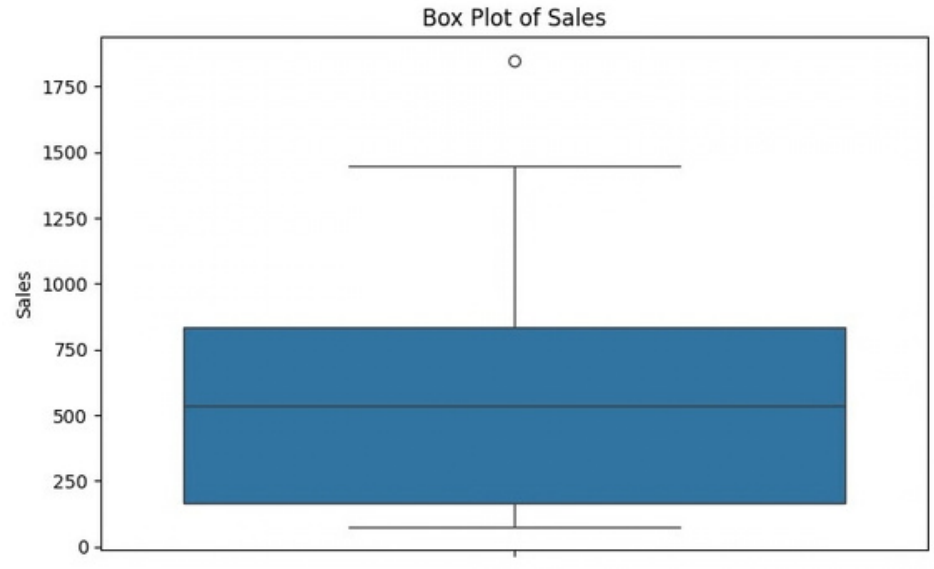
```
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

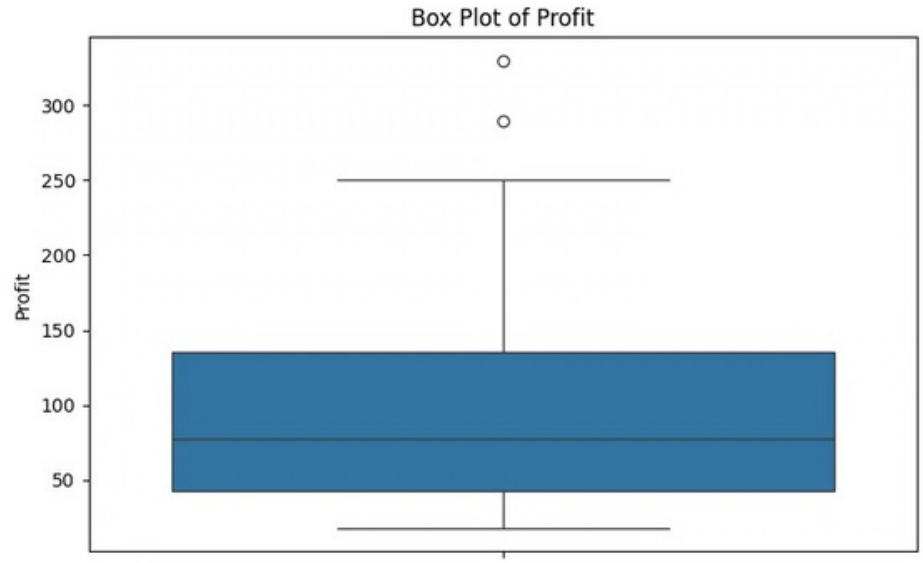
plt.title("Correlation Heatmap")
plt.show()
```




```
plt.figure(figsize=(8,      5))
sns.boxplot(y=df['Sales'])
plt.title("Box Plot of Sales")
plt.ylabel("Sales")
plt.show()
```



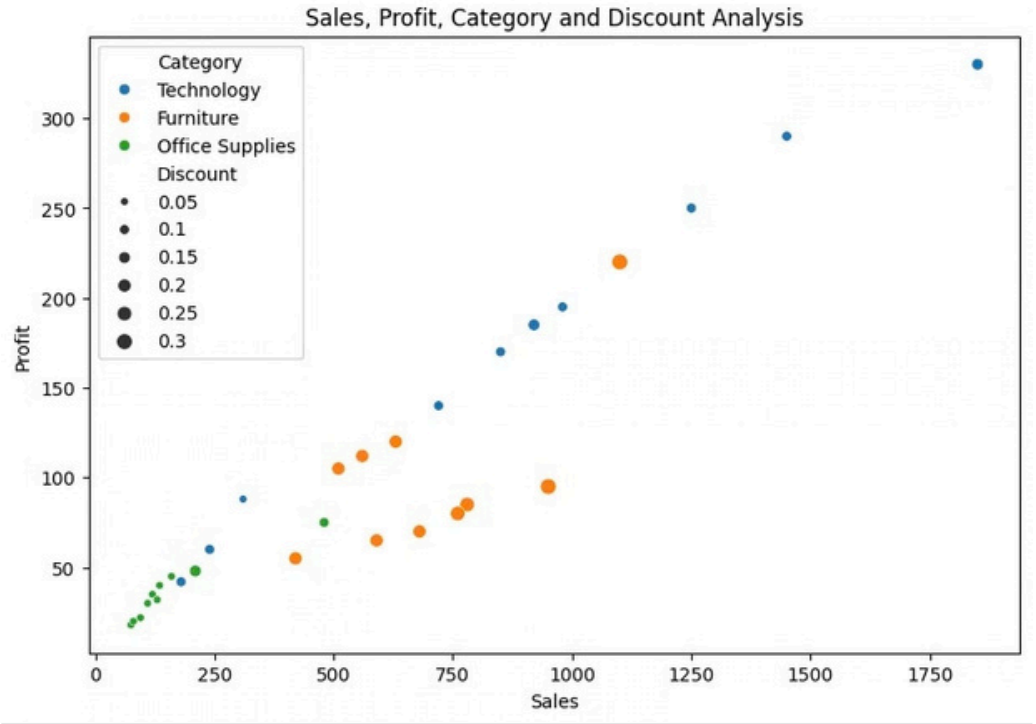

```
plt.figure(figsize=(8, 5))
sns.boxplot(y=df['Profit'])
plt.title("Box Plot of Profit")
plt.ylabel("Profit")
plt.show()
```




```
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x='Sales',
    y='Profit',
    hue='Category',
    size='Discount'
)

plt.title("Sales, Profit, Category and Discount Analysis")
plt.show()
```




```
top_products = (
    df.groupby('Product Name')['Sales']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

print("Top 10 Products by Sales:")
print(top_products)
```

```
Top 10 Products by Sales:
Product Name
Laptop Max          1850
Laptop Air          1450
```

```
Laptop  Pro  1250  Meeting
Table  1100  Smartphone  D
980
Conference Table 950
Smartphone      C      920
Smartphone  A  850  Office
Table  780  Tall  Bookcase
760
Name: Sales, dtype: int64
```

```
top_profit_products = (
    df.groupby('Product Name')['Profit']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

print("Top 10 Products by Profit:")
print(top_profit_products)
```

```
Top 10 Products by Profit:
Product Name
Laptop Max         330
Laptop Air         290
Laptop Pro         250
Meeting Table      220
Smartphone D       195
Smartphone C       185
Smartphone A       170
Smartphone B       140
Executive Chair    120
Ergonomic Chair    112
Name: Profit, dtype: int64
```

```
print("Highest Sales Category:", category_sales.idxmax())
print("Highest Profit Category:", category_profit.idxmax())
print("Highest Sales Region:", region_sales.idxmax())
```

```
Highest Sales Category: Technology
Highest Profit Category: Technology
Highest Sales Region: West
```

Questions and Answers

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

Answer: Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset using statistical methods and visualizations. It helps identify patterns, trends, relationships, outliers, missing values, and unusual observations.

EDA is performed before machine learning because it helps understand the data and identify potential problems before building a model. It also helps in selecting useful features and choosing appropriate preprocessing techniques.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Answer:

1. **Univariate Analysis:** It analyzes one variable at a time. **Example:** Studying the distribution of Sales using a histogram.
2. **Bivariate Analysis:** It analyzes the relationship between two variables. **Example:** Studying the relationship between Sales and Profit using a scatter plot.
3. **Multivariate Analysis:** It analyzes three or more variables together. **Example:** Studying Sales, Profit, Category, and Discount simultaneously.

Q3. What insights can be obtained from a correlation heatmap?

Answer: A correlation heatmap shows the strength and direction of relationships between numerical variables. Correlation values generally range from -1 to +1.

- A value close to **+1** indicates a strong positive relationship.
- A value close to **-1** indicates a strong negative relationship.
- A value close to **0** indicates a weak or no linear relationship.

It can help identify strongly related features and possible redundant variables.

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Answer:

- **Histogram:** Shows the distribution and frequency of numerical data. It helps understand the spread and shape of the data.
- **Box Plot:** Shows the median, quartiles, spread, and potential outliers in numerical data.
- **Scatter Plot:** Shows the relationship between two numerical variables and helps identify trends, patterns, and unusual obser vations.

Q5. How can EDA help identify data quality issues before analysis?

Answer: EDA helps identify data quality problems such as missing values, duplicate records, incorrect data types, unusual values, outliers, and inconsistent categories.

Functions such as info(), isnull(), describe(), and value_counts() along with graphs help analysts detect these problems before performing further analysis.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Answer: Correlation is important because it helps identify the strength and direction of relationships between variables. It can help in feature selection and understanding which variables may be useful for prediction.

However, **correlation does not imply causation**. Two variables may be correlated without one causing the other. A third factor or coincidence may explain the relationship.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

Answer: For **categorical variables**, bar charts and count plots are useful because they show the frequency or count of different categories.

For **numerical variables**, histograms and box plots are useful because they show the distribution, spread, central tendency, and outliers.
For relationships between two numerical variables, a **scatter plot** is suitable.

Q8. What business insights can be derived from the Superstore dataset through EDA?

Answer: EDA of the Superstore dataset can provide several useful business insights, such as:

1. Which category generates the highest sales.